String Theory Seminar

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Abstract

This is the lecture note of the String Theory Seminar in EPFL 2026 spring semester. The main reference will be Polchinski's String Theory. Metric convention $(-, +, +, +)$

Contents

1 Preliminary: Functional Formalism

In this section, we will discuss a functional formulation of quantum theory, which is the basis of the path integral approach to quantum field theory. The contents mainly follow the book [1] and [2].

In quantum field theory we mainly care about the transition amplitude between states and with operators inserted, which are called the correlation functions. Among all correlation functions, we mainly care about transition amplitudes between "vaccum states" with operators inserted:

$$\langle\Omega(\infty)|\mathcal{T}O(x_1)...O(x_n)|\Omega(-\infty)\rangle \tag{1}$$

It is proven that the correlation functions are related to the scattering amplitudes of particles, through the LSZ reduction formula. Therefore, the correlation functions are one of the main objects of interest in quantum field theory. (note that $|\Omega(t)\rangle$ becomes the time independent vaccum state $|0\rangle$ for a free theory)

Apart from the canonical quantization approach, there exist another approach in quantum theory, which makes the calculation of correlation functions more convenient. This approach is called the path integral approach, which is the main topic of this lecture.

1.1 Quantum Mechanics Path Integral

To formulate the path integral approach, we start with one dimensional quantum mechanic system with a standard non-relativistic Hamiltonian:

$$H = \frac{p^2}{2m} + V(x) \tag{2}$$

and focus on the standard transition amplitude between two position eigen states, then we generalize it to case with operator insertions and transition amplitude between vaccum states, and finally we will generalize it to quantum field theory.

1.1.1 Transition Amplitude as Functional Integral

We try to calculate the transition amplitude between two position eigen states $|x\rangle$ and $|x'\rangle$, by definition the transition amplitude is:

- **Schrodinger Picture :** with $|x\rangle$ as eigen state of position operator, we have:

$$\langle x'|e^{-iHt}|x\rangle \tag{3}$$

- **Heisenberg Picture :** with $|x,t\rangle$ as eigen state of position operator $x(t)$ at time t, we have:

$$\langle x',t|x,0\rangle \tag{4}$$

These two definitions are equivalent, its only a matter of choice of picture.

> **Remark**
>
> In fact we can prove that $|x,t\rangle = e^{iHt}|x\rangle$, thus directly see the equivalence of the two definitions.

The discussion in this subsection is straightforward in schrodinger picture, thus we will mainly work in schrodinger picture in this subsection.

We want to transform this operator expession into a functional integral, thus we need to insert complete set of states to make operators into c-numbers. The trick is as follows:

- Step 1: devide time interval into N small intervals $t = \sum_{i=1}^{N} \delta t$ and then insert complete set of states in each small interval:

$$\begin{aligned}\langle x'|e^{-iHt}|x\rangle &= \langle x'|e^{-iH\delta t}e^{-iH\delta t}...e^{-iH\delta t}|x\rangle \\ &= \int \prod_{i=1}^{N-1} dx_i \langle x'|e^{-iH\delta t}|x_{N-1}\rangle\langle x_{N-1}|e^{-iH\delta t}|x_{N-2}\rangle...\langle x_1|e^{-iH\delta t}|x\rangle\end{aligned} \tag{5}$$

- Step 2: evaluate each infinitesimal transition amplitude $\langle x_{i+1}|e^{-iH\delta t}|x_i\rangle$ we notice that:

$$e^{-iH\delta t} = e^{-i\frac{p^2}{2m}\delta t}e^{-iV(x)\delta t}e^{O(\delta t^2)} \tag{6}$$

> **Remark**
>
> The above formula is a consequence of the Baker-Campbell-Hausdorff formula, which states that for two operators A and B, we have:
>
> $$e^{A+B} = e^A e^B e^{-\frac{1}{2}[A,B]}... \tag{7}$$

Then we insert complete set of momentum eigen states $|p\rangle$:

$$\begin{aligned}\langle x_{i+1}|e^{-iH\delta t}|x_i\rangle &= \int \frac{dp_i}{2\pi}\langle x_{i+1}|e^{-i\delta t\frac{p^2}{2m}}|p_i\rangle\langle p_i|e^{-i\delta tV(x)}|x_i\rangle \\ &= \int \frac{dp_i}{2\pi}\exp\left[ip_i(x_{i+1}-x_i) - i\left(\frac{p_i^2}{2m}+V(x_i)\right)\delta t\right]\end{aligned} \tag{8}$$

If we do this for each term in Equation 5, we will get:

$$\langle x'|e^{-iHt}|x\rangle = \int_{x(0)=x}^{x(t)=x'}\left(\prod_{i=1}^{N-1}dx_i\prod_{j=0}^{N-1}\frac{dp_j}{2\pi}\right)\exp\left[i\sum_{i=0}^{N-1}\left(p_i(x_{i+1}-x_i)-\left(\frac{p_i^2}{2m}+V(x_i)\right)\delta t\right)\right] \tag{9}$$

- Step 3: then we integrate out p_i. This is a gaussian integral, thus we can get:

$$\langle x'|e^{-iHt}|x\rangle = \int_{x(0)=x}^{x(t)=x'}\left(\prod_{i=1}^{N-1}dx_i f(\delta t,m)\right)\exp\left[i\sum_{i=0}^{N-1}\left(\frac{m}{2}\left(\frac{x_{i+1}-x_i}{\delta}t\right)^2 - V(x_i)\right)\delta t\right] \tag{10}$$

where $f(\delta t, m)$ is caused by the gaussian integral.

> **Remark**
>
> In some contexts eg. Lattice field theory, we mainly don't integrate out p_i and keep the path integral formula in terms of both x and p.

- Step 4: we take the limit of $N \to \infty$ and $\delta t \to 0$, then we can get the final path integral formula:

$$\langle x'|e^{-iHt}|x\rangle = \int_{x(0)=x}^{x(t)=x'}\mathcal{D}x\exp\left[i\int_0^t dt\left(\frac{m}{2}\dot{x}^2 - V(x)\right)\right] \tag{11}$$

Where we define the measure of the path integral as:

$$\mathcal{D}x = \lim_{N\to\infty}\left(\prod_{i=1}^{N-1}dx_i f(\delta t,m)\right) \tag{12}$$

Moreover, we notice that the exponent in the path integral is just the action of the system, thus we can write the path integral formula in a more compact way:

$$\langle x'|e^{-iHt}|x\rangle = \int_{x}^{x'} \mathcal{D}x \exp(iS[x]) \tag{13}$$

> **Remark**
>
> Notice that the path integral measure is in fact a mathematically ill-defined object. Yet to make it physical meaningful, we have to impose some physical conditions on it.

1.1.2 Correlation Functions as Functional Integral

Correlation functions are transition amplitudes with time ordered operator insertions, in Heisenberg picture, we can write the correlation functions as:

$$\langle x', t'|\mathcal{T}O(t_1)...O(t_n)|x, t\rangle \tag{14}$$

where $|x, t\rangle$ is the eigen state of position operator $x(t)$ at time t. By repeating the same procedure as in the previous subsection, we can get the path integral formula for the correlation functions:

$$\langle x', t'|\mathcal{T}O(t_1)...O(t_n)|x, t\rangle = \int_{x}^{x'} \mathcal{D}xO(t_1)...O(t_n) \exp(iS[x]) \tag{15}$$

where in the LHS, $O(t)$ are operators, while in the RHS, $O(t)$ are c-numbers.

1.1.3 Correlation Functions with Vaccum States

We mainly care about the correlation functions between vaccum states at $-\infty$ and ∞ time (while for a free theory the vaccum state is independent of time, for a interacting theory we can define sth as the vaccum state in infinity time which annilated by $a(\infty)$), which are defined as:

$$\langle 0|\mathcal{T}O(t_1)...O(t_n)|0\rangle \tag{16}$$

> **Remark**
>
> For simplicity, we sometimes write the correlation function between vaccum states as $\langle O(t_1)...O(t_n)\rangle$

We can use a trick to transform this correlation function into something easier to calculate.

- An Important Fact:

We notice that if we normalize the Hamiltonian to have the vaccum energy to be zero, then the vaccum state can be generated by:

$$|0\rangle \propto \lim_{T\to\infty} e^{-HT}|\psi\rangle \tag{17}$$

This is because as $T \to \infty$, the contribution from the excited states will be exponentially suppressed only the vaccum state will survive as long as $|\psi\rangle$ has non-zero overlap with the vaccum state.

- Trick:

We modify the Hamiltonian by adding a small imaginary part:

$$H \to (1 - i\varepsilon)H \quad \varepsilon > 0 \tag{18}$$

Then we notice that consider an arbitrary state $|\psi\rangle, |\psi'\rangle$ with non-zero overlap with the vaccum state, we have:

$$\lim_{\substack{t'\to\infty \\ t\to-\infty}} \langle\psi'|e^{-i(1-i\varepsilon)H(t'-t)}\ |\psi\rangle \propto \lim_{\substack{t'\to\infty \\ t\to-\infty}} \langle 0|e^{-iH(t'-t)}|0\rangle \tag{19}$$

Thus we have:

$$\langle 0|0\rangle \propto \int \mathcal{D}x e^{iS_\varepsilon[x]} \tag{20}$$

- We don't write out the boundary condition for its arbitrary as long as we find suitable normalization of the measure.
- $S_\varepsilon[x]$ is the associate action with the modified Hamiltonian.

Similarly, for the correlation functions, we have:

$$\langle O(t_1)...O(t_n)\rangle := \langle 0|\mathcal{T}O(t_1)...O(t_n)|0\rangle \propto \int \mathcal{D}xO(t_1)...O(t_n)\exp(iS_\varepsilon[x]) \tag{21}$$

> **Remark**
>
> We write proportional to instead of equal to because in Equation 17 taking limit may cause some subtleties, yet as long as we keep some suitable normalization, we can get rid of the subtleties and get a rational result.

1.1.4 Generating Functional and Functional Derivative

Generating functional is a very useful tool to calculate correlation functions in quantum theory. We define a generating functional as the following functional integral:

$$Z[J] = \int \mathcal{D}x \exp\left(iS_\varepsilon[x] + i\int dtJ(t)x(t)\right) \tag{22}$$

Then we can see that the correlation functions can be calculated by taking functional derivatives of the generating functional:

$$\begin{aligned} \langle x(t_1)...x(t_n)\rangle &\propto \int \mathcal{D}xx(t_1)...x(t_n)\exp\left(iS_\varepsilon[x] + i\int dtJ(t)x(t)\right)|_{J=0} \\ &= (-i)^n \frac{\delta^n}{\delta J(t_1)...\delta J(t_n)} Z[J]|_{J=0} \end{aligned} \tag{23}$$

Now we can discuss about the normalization. For free theory we can see that $Z[0] \propto \langle 0|0\rangle = 1$, thus long as we choose a normalization with $Z[0] = 1$ we can make the $\propto$ an equal sign in all above formula. For a free theory $Z[J]$ can be explicitly calculated, after the normalization is fixed, in the following part of this note, we will use the normalization with $Z[0] = 1$.

1.1.5 Euclidian path integral

Now we make an assumption that:

- The correlation function $\langle x(t_1)...x(t_n)\rangle$ as a function of time $C(t_1, ...t_n)$ can be analyticaly continued from real time to imaginary time.

Then we can make an analytical continuation of the correlation function from real time to imaginary time $t = -i\tau, \tau \in \mathbb{R}$, which is called the Euclidian correlation function:

$$\langle x_E(\tau_1)...x_E(\tau_n)\rangle := C(-i\tau_1, ... - i\tau_n) \tag{24}$$

Then due to the relation between the correlation function and the functional integral Equation 21, we can also get a functional integral formula for the Euclidian correlation function(The RHS is called the Wick Rotation of the original path integral):

$$\langle x_E(\tau_1)...x_E(\tau_n)\rangle = \int \mathcal{D}x_E x_E(\tau_1)...x_E(\tau_n)\exp(-S_E[x_E]) \tag{25}$$

Where $x_E(\tau) \coloneqq x(-i\tau) = x(t)$ and $S_E[x_E] = -iS[x]$ is the Euclidian Action, we define it like this for simplicity. For example, a quantum mechanical system:

$$S_E[x_E] = \int d\tau\Big(\frac{m}{2}(\partial_\tau x_E)^2 + V(x_E)\Big) \tag{26}$$

In CFT we mainly care about Euclidian Path Integral of Euclidian Action on Euclidian Space. If we want to get a correlation function of the related Lorentzian theory, we just do an analytical continuation back by taking the result function from $\tau \in \mathbb{R}$ to $\tau = it, t \in \mathbb{R}$.

> **Remark**
>
> We no more have the ε in the Euclidian path integral. In fact we can view $H \to (1 - i\varepsilon)H$ equivalently as $t \to (1 - i\varepsilon)t$ in the path integral. We are just considering an extreme case where $-i\varepsilon t$ dominantes.

1.2 Free Scalar Field Path Integral

Now we generalize the path integral fomalism from quantum mechanics to quantum field theory. We will mainly discuss about the path integral of free scalar fields. In this section, we will mainly use the Lorentzian path integral. The action of a free real scalar field is:

$$S = \int d^4x\Big(-\frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi - \frac{1}{2}m^2\varphi^2\Big) \tag{27}$$

1.2.1 Generalize from QM

Path Integral of QFT can be obtained from the path integral of QM simply by doing:

- $x(t) \to \varphi_i(x, t)$
- $J(t) \to J(x, t)$

And we integrate over space time in the action term and integrate over all field configuration in the path integral.

$$Z[J] = \int \mathcal{D}\varphi \exp\Big(iS_\varepsilon[\varphi] + i\int d^4x J(x)\varphi(x)\Big) \tag{28}$$

and the correlation functions can be calculated by taking functional derivatives of the generating functional:

$$\begin{aligned}\langle\varphi(x_1)...\varphi(x_n)\rangle &= \int \mathcal{D}\varphi\varphi(x_1)...\varphi(x_n)\exp\Big(iS_\varepsilon[\varphi] + i\int d^4x J(x)\varphi(x)\Big)|_{J=0} \\ &= (-i)^n \frac{\delta^n}{\delta J(x_1)...\delta J(x_n)} Z[J]|_{J=0}\end{aligned} \tag{29}$$

1.2.2 Exact Generating Functional

Generating functional of a free theory can be exactly calculated. We first look at the action:

$$S_\varepsilon[\varphi] + \int d^4x J(x)\varphi(x) = \int d^4x\Big(-\frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi - \frac{1}{2}(m^2 - i\varepsilon')\varphi^2 + J\varphi\Big) \tag{30}$$

notice that we use a trick of taking $-i\varepsilon H$ as a correction to the mass term: $-i\varepsilon H = -\frac{1}{2}i\varepsilon'\varphi^2$. Then the action is just the standard action with a imaginary shift to the mass, in the following we will just write m^2 but really means $m^2 - i\varepsilon'$.

After an integration by part, we can rewrite the action as:

$$\begin{aligned} S_\varepsilon[\varphi] + \int d^4x J(x)\varphi(x) &= -\frac{1}{2}\int d^4x\varphi(-\partial^2 - m^2)\varphi + \int d^4x J\varphi \\ &= \int d^4x\Big(-\frac{1}{2}\varphi M\varphi + J\varphi\Big) \end{aligned} \tag{31}$$

where $M = -\partial^2 - m^2$ is a linear operator. We know that how to do integral with this stuff on the exponential, see that:

$$\int d^n x \exp\Big(-\frac{1}{2}x^T M x - J^T x\Big) = (2\pi)^{n/2}(\det\ M)^{-1/2}\exp\Big(\frac{1}{2}J^T M^{-1} J\Big) \tag{32}$$

We generalize this into a functional case with a imaginary i ahead, then we can get:

$$Z[J] = \exp\Big(\frac{i}{2}\int d^4x d^4y J(x) D_F(x-y) J(y)\Big) Z[0] \tag{33}$$

where $D_F(x-y)$ can be viewed as the inverse of the operator M, satisfying the following equation:

$$(-\partial^2 - m^2) D_F(x-y) = \delta^4(x-y) \tag{34}$$

and can be explicitly written as:

$$D_F(x-y) = \int \frac{d^4p}{(2\pi)^4}\frac{\exp(-ip(x-y))}{p^2 + m^2 - i\varepsilon} \tag{35}$$

this turns out to be exactly the Feymann propagator of the free scalar field.

> **Remark**
>
> Surely, this is not a rigorous derivation, but a fast way to believe the result is correct.

1.2.3 Calculation of Correlation Functions

Now we can calculate the correlation functions of the free scalar field by taking functional derivatives of the generating functional. For example, the two point correlation function is:

$$\langle\varphi(x)\varphi(y)\rangle = -i\frac{\delta^2}{\delta J(x)\delta J(y)} Z[J]|_{J=0} = -iD_F(x-y) \tag{36}$$

We can see that indeed $D_F(x-y)$ is the Feymann propagator of the free scalar field.

Here we can easily see the Wick's theorem for free theory, that the correlation function can be written as a sum of products of two point correlation functions. This is because the generating functional is a exponential of a quadratic term of J, thus when we take functional derivatives, we will get a sum of products of two point correlation functions.

1.3 Quantum Conservation and Ward identity

Classically, Noether's theorem states that for a system with a continuous symmetry, there is a conserved current J^μ and a conserved charge Q. In a quantum theory, we can find that the symmetry will lead to constraints on the correlation functions, which are called the Ward identities. In this section, we will discuss about this using a Euclidian path integral.

1.3.1 Noether's Theorem

We define a symmetry transformation in the following way:

Definition 1.1 **Global Symmetry Transformation**

A symmetry transformation is a coordinate transformation with a constant parameter ω :

$$x \to x'^\mu = f^\mu(x, \omega) \quad \varphi_i(x) \to \varphi_i'(x') = F_i(\varphi(x), \omega) \tag{37}$$

$$\varphi_i'(x) = F_i(\varphi(f^{-1}(x)), \omega) \quad \delta\varphi_i(x) = \varphi_i'(x) - \varphi_i(x) = -i\omega_a G_a \varphi_i(x) \tag{38}$$

Such that the action is invariant under this transformation:

$$S[\varphi'(x)] = S[\varphi(x)] \quad \text{or infinitesimally} \quad \delta S = 0 \tag{39}$$

With this definition, if we lift the parameter ω_a to depend on the spacetime $\omega_a(x)$. Then the viration of the action will not be zero, but will be propotional to the derivative of $\omega_a(x)$. We can write the viration of the action as:

$$\delta S = \int d^4x - J_a^\mu \partial_\mu \omega_a(x) = \int d^4x (\partial_\mu J_a^\mu) \omega_a(x) \tag{40}$$

Thus if we consider on-shell condition, that under any viration of the field, the action is invariant, then we can get the classical conservation law:

$$\partial_\mu J_a^\mu = 0 \tag{41}$$

This is known as the Noether's theorem.

1.3.2 Coorelation Functions under Symmetry Transformation

For a quantum theory, we not only need the action to be invariant under the symmetry transformation, but also need the path integral measure to be invariant under the symmetry transformation $\mathcal{D}\varphi' = \mathcal{D}\varphi$. Then we consider the following Euclidian correlation function:

$$\langle \varphi(x_1')...\varphi(x_n') \rangle = \int \mathcal{D}\varphi \varphi(x_1')...\varphi(x_n') \exp(-S[\varphi]) \tag{42}$$

We simple rename the integral variable from $\varphi(x)$ to $\varphi'(x)$, then we can get:

$$\langle \varphi(x_1')...\varphi(x_n') \rangle = \int \mathcal{D}\varphi' \varphi'(x_1')...\varphi'(x_n') \exp(-S[\varphi']) \tag{43}$$

We then use the two symmetry conditions to transform the measure and the action $\mathcal{D}\varphi' = \mathcal{D}\varphi$ and $S[\varphi'] = S[\varphi]$, then we can get:

$$\langle \varphi(x_1')...\varphi(x_n') \rangle = \int \mathcal{D}\varphi \varphi'(x_1')...\varphi'(x_n') \exp(-S[\varphi]) = \langle \varphi'(x_1')...\varphi'(x_n') \rangle \tag{44}$$

Thus we can see that the correlation function should satisfy following relation:

$$\begin{gathered} \langle \varphi'(x_1)...\varphi'(x_n) \rangle = \langle \varphi(x_1)...\varphi(x_n) \rangle \\ \text{or equivalently } \langle F(\varphi(x_1), \omega)...F(\varphi(x_n), \omega) \rangle = \langle \varphi(x_1')...\varphi(x_n') \rangle \end{gathered} \tag{45}$$

1.3.3 Ward Identity

We can also derive another type of constraint infinitesimally. We consider a local transformation by lifting the parameter ω_a to depend on the spacetime $\omega_a(x)$. Then we notice that:

$$\langle\varphi(x_1)...\varphi(x_n)\rangle = \int \mathcal{D}\varphi' \varphi'(x_1)...\varphi'(x_n) \exp(-S[\varphi']) \tag{46}$$

We consider to expand the RHS to the first order of $\omega_a(x)$, now:

$$\varphi'(x_i) = \varphi(x_i) - i\omega_a(x_i) G_a \varphi(x_i) \tag{47}$$

$$S[\varphi'] = S[\varphi] + \delta S = S[\varphi] + \int d^4x (\partial_\mu J_a^\mu) \omega_a(x) \tag{48}$$

Then we can expand the RHS of Equation 46 to the first order of $\omega_a(x)$, we can get:

$$\begin{aligned}
&\langle\varphi(x_1)...\varphi(x_n)\rangle \\
=& \int \mathcal{D}\varphi(\varphi(x_1) - i\omega_a(x_1)G_a\varphi(x_1))...(\varphi(x_n) - i\omega_a(x_n)G_a\varphi(x_n)) \exp\left(-S[\varphi] - \int d^4x(\partial_\mu J_a^\mu)\omega_a(x)\right) \\
=& \int \mathcal{D}\varphi\varphi(x_1)...\varphi(x_n) \exp(-S[\varphi]) + i\int d^4x \omega_a(x) \int \mathcal{D}\varphi(\partial_\mu J_a^\mu)\varphi(x_1)...\varphi(x_n) \exp(-S[\varphi]) \\
&-i\sum_{i=1}^{n} \omega_a(x_i) \int \mathcal{D}\varphi\varphi(x_1)...G_a\varphi(x_i)...\varphi(x_n) \exp(-S[\varphi])
\end{aligned} \tag{49}$$

Finally, due to the arbitrary choice of $\omega_a(x)$, we compare the integrant, and this leads to the Ward identity:

$$\partial_\mu \langle J_a^\mu(x)\varphi(x_1)...\varphi(x_n)\rangle = -i\sum_{i=1}^{n} \delta^4(x - x_i)\langle\varphi(x_1)...G_a\varphi(x_i)...\varphi(x_n)\rangle \tag{50}$$

Bibliography

[1] M. A. Srednicki, *Quantum Field Theory*. Cambridge ; New York: Cambridge University Press, 2007.

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